NATHANIEL McINTIRE

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PROFESSIONAL SUMMARY

Senior aerospace engineering student at NC State focused on structures, materials, and aerodynamics. Experience in MATLAB, ANSYS, and hands-on rover development, with leadership as Payload and Test/Safety Lead for a lunar rover senior design project.

EDUCATION

NORTH CAROLINA STATE UNIVERSITY **Expected Graduation: May 2027**

Bachelor of Aerospace Engineering | *Minors: Astrophysics, Physics, Mathematics*

- Relevant Coursework: Aerodynamics I & II, Aerospace Structures I & II, Automatic Controls, Space Flight, Thermodynamics, Numerical Analysis, FEA, Rocket Propulsion, MATLAB

BRUNSWICK COMMUNITY COLLEGE **May 2024**

Associate of Engineering | *Associate of Science* | *Cum Laude Honors*

ENGINEERING PROJECTS & DESIGN EXPERIENCE

NASA RASC-AL LUNAR RESOURCE EXPLORATION ROVER - PAYLOAD LEAD **August 2026 - Present**

- Developing a lunar rover concept to map permanently shadowed regions (PSRs) and characterize volatile and mineral resources found in regolith.
- Lead payload development, including the regolith breakup mechanism, selection of resource-analysis instruments, and other sensors needed for operation.
- Coordinate sensor and mechanism requirements with other subteam leads to integrate the payload into the rover design.
- Selected LiDAR for terrain mapping in PSRs, where navigation must operate without sunlight.
- Selected XRF, near-infrared, and mass spectrometry instruments to measure regolith composition and volatile resources.

FAUX FLIGHT — MICRO-LARP ROVER PROTOTYPE **October 2026**

- Electrical integration and software development, working with a teammate to connect and control the rover's motors and sensors.
- Integrated a Raspberry Pi Pico 2 W, motor drivers, scanning LiDAR, and temperature-controlled cooling into a four-wheel-drive rover.
- Collaborated on Python software for wireless laptop control, sensor telemetry, and motor operation.
- Troubleshot power delivery and wireless connectivity during testing; the team completed all ten demonstration requirements within a $200 budget.

COMPOSITE LAMINATE STRESS ANALYSIS **May 2026**

- Evaluated ply-level stresses in composite laminates under multiple loading conditions and transformed stresses between material and global coordinate systems.
- Used MATLAB and classical laminate theory to compare carbon/epoxy stacking sequences, calculate bending stiffness, and predict first-ply failure using the maximum stress criterion.

CONVERGING-DIVERGING NOZZLE DESIGN AND FLOW ANALYSIS **December 2025**

- Designed a converging-diverging nozzle in SolidWorks, manufactured it via 3D printing, and modeled internal pressure and velocity in ANSYS CFD.
- Conducted wind-tunnel testing at NC State and processed experimental data in MATLAB to compare measured and computational flow characteristics.

TECHNICAL SKILLS

Software: MATLAB | Python | C++ | ANSYS | SolidWorks | XFOIL | Excel | KiCad

Engineering Methods: FEA | CFD | Structural Analysis | Composite Stress Analysis | Control System Design | PLC Programming

PROFESSIONAL EXPERIENCE

PRANZO ITALIANO | Server | Raleigh, NC **September 2026 - Present**

BONGIORNO & SON ITALIAN SPECIALTIES | Supervisor | Raleigh, NC **March 2025 - August 2026**

ACE HARDWARE | Supervisor | Raleigh, NC **August 2024 - March 2025**

LOWES FOODS | Guest Services Manager | Wilmington, NC **October 2020 - August 2024**